Roll No.:

Test Date: 24-11-2024

Aakash
Medical | IIT-JEE | Foundations

F
Booklet Code

AR-008

AIATS
ALL INDIA AAKASH TEST SERIES

for

Medical Entrance Exam - 2025

National Eligibility-cum-Entrance Test (NEET)

TEST No. 2

(XII Passed Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.
9. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
10. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
11. There are two sections in each subject i.e., Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions out of 15 from Section-B.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 2

TOPICS OF THE TEST

Physics	Laws of Motion; Work, Energy and Power
Chemistry	Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Botany	Biological Classification, Morphology of Flowering Plants
Zoology	Breathing and Exchange of Gases, Body Fluids and Circulation

MM : 720

TEST - 2

Time : 3 Hrs. 20 Min.

[PHYSICS]

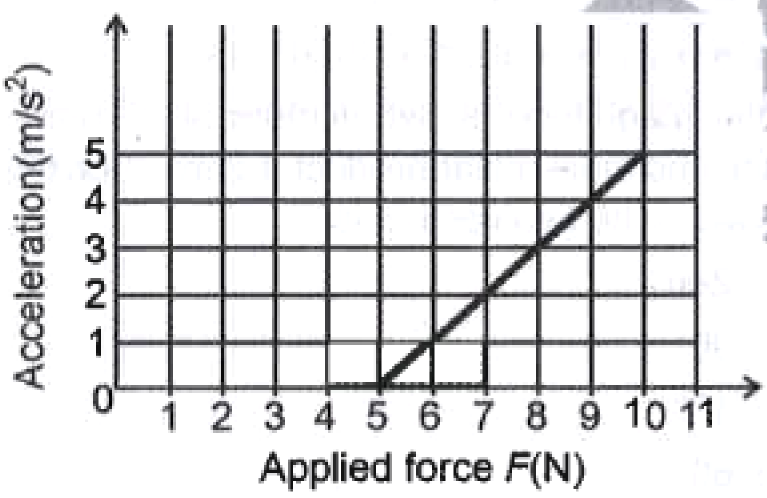
Choose the correct answer :

SECTION-A

1. A bend in a level road has a radius of 5 m. The maximum speed with which a car can turn along the bend without skidding is (Assume the coefficient of friction to be 0.5)
   (1) 2 m/s (2) 5 m/s
   (3) 10 m/s (4) 8 m/s

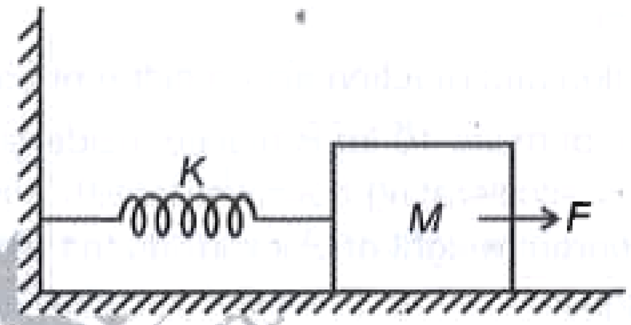
2. If a block moving in gravity-free space region breaks into two parts, then
   (1) The total linear momentum of system must remain conserved.
   (2) The total kinetic energy of system must be conserved.
   (3) The total linear momentum of system must change.
   (4) The linear momentum of individual parts must remain conserved.

3. A force F is applied on a block of mass m which is initially at rest on a rough horizontal surface. If acceleration of the block with respect to the applied force is as shown in graph, then the value of m is



   (1) $m = 2$ kg
   (2) $m = 3$ kg
   (3) $m = 5$ kg
   (4) $m = 1$ kg

4. A block of mass M is resting on a smooth horizontal surface. It is connected to a rigid wall by a massless spring of stiffness K. The spring is in its natural length.



If a constant horizontal force starts acting on the block towards right, then the extension in spring when block is in equilibrium position will be
   (1) $\frac{F}{K}$ (2) $\frac{2F}{K}$
   (3) Zero (4) $\frac{F}{2K}$

5. Consider a car and a bus of mass M_1 and M_2 respectively. If they are moving with same velocity, then the ratio of their stopping distance under the influence of same retarding force will be
   (1) $\frac{M_1}{M_2}$ (2) 1
   (3) $\frac{M_1^2}{M_2^2}$ (4) $\frac{M_1}{2M_2}$

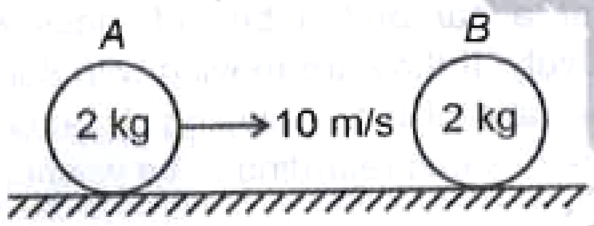
Space for Rough Work

6. Which among the following options represent impulse?
   (1) Mass × acceleration
   (2) Force × time
   (3) Rate of change of linear momentum
   (4) Rate of change in potential energy

7. Choose the correct statement with regards to Newton's third law of motion.
   (1) To every action, there may not be an equal and opposite reaction.
   (2) Action and reaction forces have a cause-effect relationship.
   (3) Action-reaction forces have time lag between them.
   (4) Action and reaction act on different bodies.

8. A block of mass 10 kg is placed inside an elevator which is accelerating downwards with 2 m/s^2, then the apparent weight of block inside the elevator is
   (1) 100 N
   (2) 20 N
   (3) 10 N
   (4) 80 N

9. Consider a head on perfectly elastic collision of two identical bodies of mass 2 kg as shown in figure. If initially A is moving with 10 m/s and B is at rest, then after the collision speed of A and B will be



   (1) V_A = 2 m/s, V_B = 8 m/s
   (2) V_A = 5 m/s, V_B = 5 m/s
   (3) V_A = 10 m/s, V_B = Zero
   (4) V_A = Zero, V_B = 10 m/s

10. Consider the following statements.
    Statement A : Coefficient of friction between any two bodies in contact depends on the nature of material of the surfaces in contact.
    Statement B : Kinetic friction is a self-adjusting force.
    Based upon above information, pick the correct option.
    (1) Statement A is true but B is false
    (2) Statement A is false but B is true
    (3) Both the statements are true
    (4) Both the statements are false

11. Column-A contains various units and column-B contain different values in other units. Match column-A and column-B and choose the correct options that follows.

Column-A	Column-B
a. 1 joule	(p) 1.6×10^{-19} joule
b. 1 eV	(q) 3.6×10^{6} joule
c. 1 kWh	(r) 10^{7} ergs
d. 1 HP	(s) 746 watt

    (1) a→ (p), b→ (r), c→ (q), d→ (s)
    (2) a→ (s), b→ (p), c→ (q), d→ (r)
    (3) a→ (r), b→ (q), c→ (s), d→ (p)
    (4) a→ (r), b→ (p), c→ (q), d→ (s)

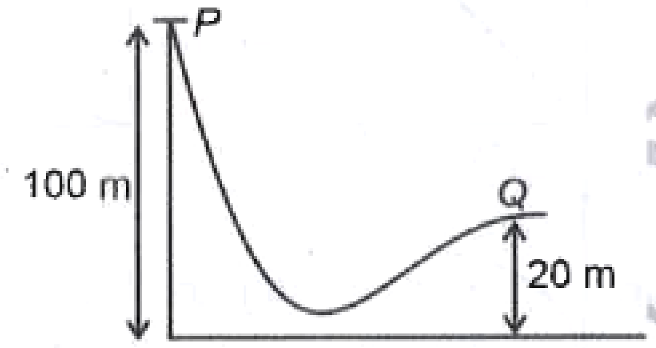
12. If the magnitude of two vectors are 7 units and 5 units and their scalar product is zero, then the angle between the two vectors is
    (1) Zero
    (2) 90°
    (3) 45°
    (4) 60°

Space for Rough Work

13. Potential energy associated with a force is given as $U = (x^2 - 5x)$ J (where x is in meter). The magnitude of force at $x = 1$ m will be
   (1) 1 N (2) 7 N
   (3) 4 N (4) 3 N

14. A sharp stone of mass 1 kg is dropped from 10 m high building. If after entering the ground to a depth of 2 m the stone comes to rest, then the average resistance force offered by ground to the stone is
   (1) 20 N
   (2) 30 N
   (3) 50 N
   (4) 60 N

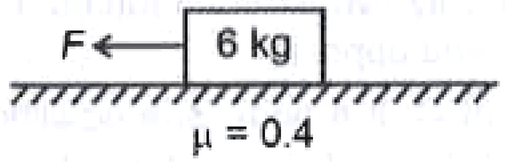
15. A body is released from position P as shown in figure. The speed of body at position Q will be [Assume surface to be smooth]



   (1) $10\sqrt{2}$ m/s
   (2) $2\sqrt{10}$ m/s
   (3) 40 m/s
   (4) 20 m/s

16. Choose the correct relation among the following for vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$.
   (1) $\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A}$
   (2) $\vec{A}\cdot(\vec{B}+\vec{C}) = \vec{A}\cdot\vec{B} + \vec{A}\cdot\vec{C}$
   (3) $\vec{A}\cdot\vec{A} = A$
   (4) Both (1) and (2)

17. Consider a block of mass 6 kg placed on a horizontal table. If the coefficient of static friction between the block and table is 0.4 and a horizontal force F is applied on the block as shown. The minimum value of F to just move the block will be



   (1) 12 N (2) 20 N
   (3) 24 N (4) 18 N

18. Choose the incorrect statement concerning inertial frame of reference.
   (1) An inertial frame of reference is an accelerated frame.
   (2) An inertial frame of reference is an unaccelerated frame.
   (3) An elevator moving with constant velocity is an example of inertial frame.
   (4) Newton's laws of motion are valid in inertial frame of reference.

19. The SI unit of power is
   (1) joule (2) newton
   (3) watt (4) erg

20. For a simple pendulum of length 1 m and mass of bob 2 kg has maximum angular displacement of $\frac{\pi}{3}$ rad, then its maximum kinetic energy is
   (1) 2 J (2) 5 J
   (3) 10 J (4) 25 J

21. If a force $\vec{F} = \left(2\hat{i} + 3\hat{j}\right)$ N acts on a body for 10 s and displacement of the body is given by $\vec{d} = 2\hat{k}$ m. Work done by this force on the body is
   (1) 4 J (2) 6 J
   (3) 10 J (4) Zero

Space for Rough Work

22. Consider the following statements

Statement-A: If a body is acted upon by a single external force, it cannot be in equilibrium.

Statement-B: If a body is in equilibrium under the action of only two external forces, the forces must be equal and opposite.

Statement-C: If a body is in equilibrium under the action of three forces, then they may not be coplanar.

Choose the correct option.

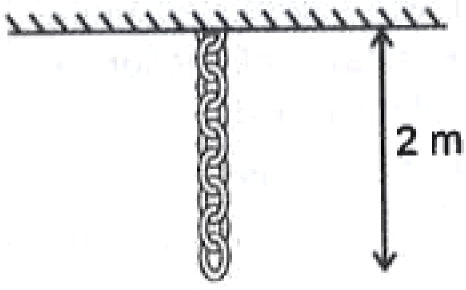
(1) Only statement A is correct

(2) All A, B and C are correct

(3) Both A and C are correct

(4) Both A and B are correct

23. A chain of mass 10 kg and length 2 m is hung vertically from the ceiling as shown in figure. The tension in the chain at 0.5 m distance from ceiling is



(1) 25 N

(2) 100 N

(3) 150 N

(4) 75 N

24. A particle is subjected to a force of 10 N and the displacement of the particle in the direction of force varies as $x = 5t - 12$, where x is in metre and t is in seconds. The value of instantaneous power at $t = 2$ s will be

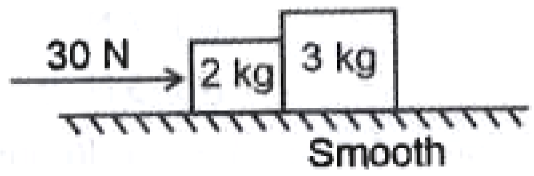
(1) 20 W

(2) 50 W

(3) 10 W

(4) 12 W

25. Consider the following diagram where a force of 30 N acts on a 2 kg block. The contact force between 2 kg block and 3 kg block will be



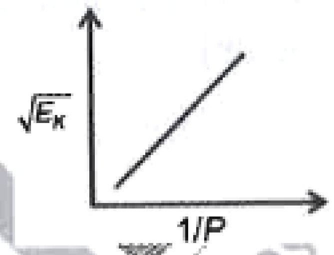
(1) 20 N

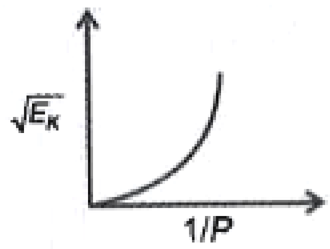
(2) 30 N

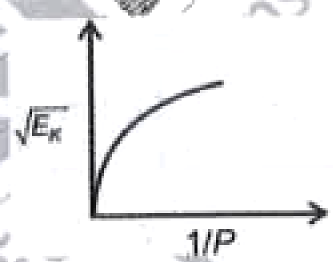
(3) 18 N

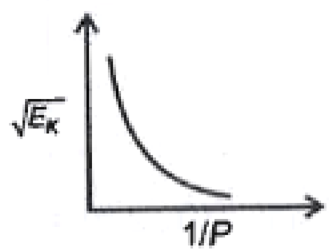
(4) 16 N

26. If E_K represents the kinetic energy of a body and P is its linear momentum, then the correct graph between $\sqrt{E_K}$ and $\frac{1}{P}$ is

(1) 

(2) 

(3) 

(4) 

27. A ball is dropped from the tower of height 20 m. If after 3rd rebound from the ground its velocity is observed to be 20 m/s, then value of coefficient of restitution for the collision between ground and ball is

(1) $\frac{1}{3}$

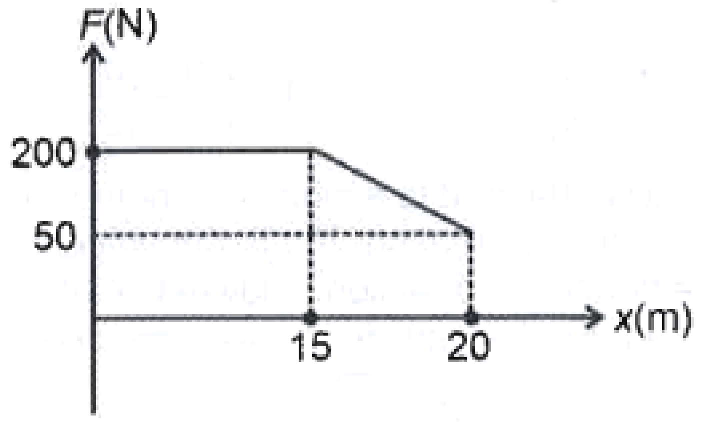
(2) $\frac{2}{3}$

(3) 1

(4) Zero

Space for Rough Work

28. A particle of mass M is subjected to a force which varies with the distance as shown.



The work done by the force during the entire process will be

(1) 1.725 kJ (2) 4 kJ
(3) 3.625 kJ (4) 4.125 kJ

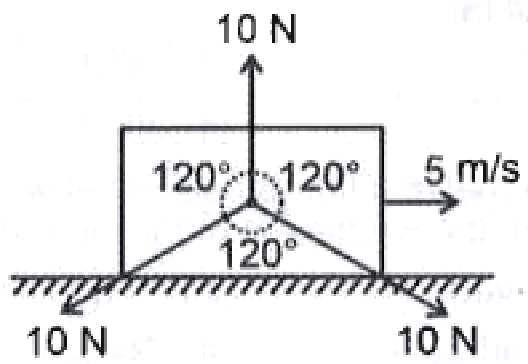
29. Consider the vectors $\vec{A} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{B} = \hat{i} + \hat{j} + \hat{k}$. The projection of $\vec{A}$ on $\vec{B}$ is

(1) $2\sqrt{2}$ (2) $\sqrt{3}$
(3) 2 (4) $2\sqrt{3}$

30. A body of mass m falls from a height h (where h is measured from ground level). The average force experienced by it before it strikes the ground is

(1) Zero (2) mg
(3) $\frac{mg}{2}$ (4) $2mg$

31. A body is acted upon by three co-planar forces of 10 N each as shown. If its initial velocity is 5 m/s in the direction shown in figure, the displacement in first 5 s is (Assume surface to be smooth).



(1) 4 m
(2) 5 m
(3) 20 m
(4) 25 m

32. Consider the following statements and choose the incorrect option.

(1) For a body of fixed mass m, $F_{net} = 0$ implies $a = 0$.
(2) The Newton's second law of motion is a vector law.
(3) The acceleration of a particle depends on history of motion of particle.
(4) If a force is not parallel to the velocity of the body, but makes some angle with it, it changes only the componeht of velocity along the force and component of velocity normal to force remains unchanged.

33. When a body is falling freely from some height, work done by gravity is

(1) Negative
(2) Positive
(3) Zero
(4) May be positive or negative

34. **Assertion (A):** For a given mass, greater the speed, the greater is the opposing force needed to stop the body in a certain time.

Reason (R): The greater the change in the momentum in a given time, the greater is the force that needs to be applied.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is false but (R) is true
(4) (A) is true but (R) is false

Space for Rough Work

35. **Statement A:** If the net external force on a body is zero, then its acceleration is non-zero.

Statement B: Acceleration will be non-zero only if there is a net external force on the body.

(1) Statement A is true but statement B is false

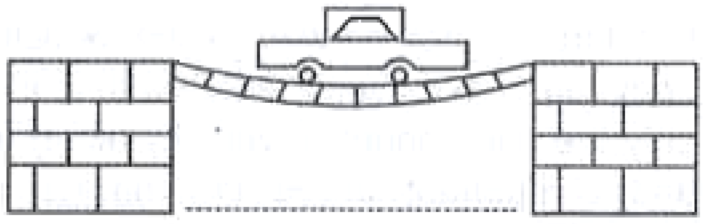
(2) Statement A is false but statement B is true

(3) Both statements A and B are true

(4) Both statements A and B are false

SECTION - B

36. For a motor car moving over a concave bridge as shown in figure, choose the correct statement.



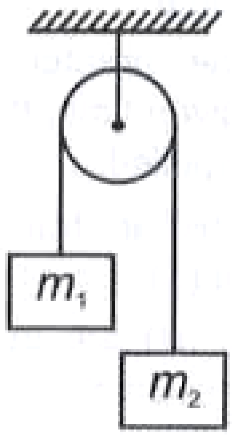
(1) Motor car moving over bridge is lighter than the same car resting on the same bridge.

(2) Motor car moving over bridge is heavier than the same car resting on the same bridge.

(3) Weight will be same in both the cases.

(4) Data insufficient

37. Two bodies of mass m_1 and m_2 are tied to the ends of a massless string. The string passes over a pulley which is frictionless.

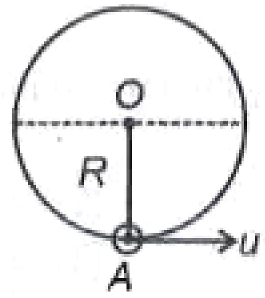


Assuming the string to be inextensible and $m_2 > m_1$, the tension in the string will be

(1) $\frac{m_1+m_2}{2m_1m_2g}$ (2) $\frac{2m_1m_2g}{m_1+m_2}$

(3) $\frac{m_1m_2g}{m_2-m_1}$ (4) $\frac{2m_1m_2g}{m_2-m_1}$

38. A particle attached to a string of length R undergoes vertical circular motion. If u is the velocity imparted in the horizontal direction at lowest point A, then the condition for the particle to leave the circle is



(1) $0 < u \le \sqrt{2gR}$ (2) $\sqrt{2gR} < u < \sqrt{5gR}$

(3) $u \ge \sqrt{5gR}$ (4) $u \ge \sqrt{7gR}$

39. An astronaut accidently gets separated out of a small spaceship accelerating in interstellar space at a constant rate of 110 m/s². The acceleration of astronaut at the instant after she is outside the spaceship is

(1) a = 10 m/s² (2) a = 110 m/s²

(3) a = Zero (4) $a = \frac{5}{3}$ m/s²

40. A body of mass 4 kg is moving under the influence of a force, due to which the variation of displacement (x) of particle with time (t) is given by $x = \frac{t^2}{2}$. The work done by the force in time interval t = 2 s to t = 4 s will be

(1) 24 J (2) 48 J

(3) 12 J (4) 36 J

41. When normal reaction is halved between two bodies in contact, the coefficient of friction between them

(1) Is halved (2) Is doubled

(3) Is quadrupled (4) Remains same

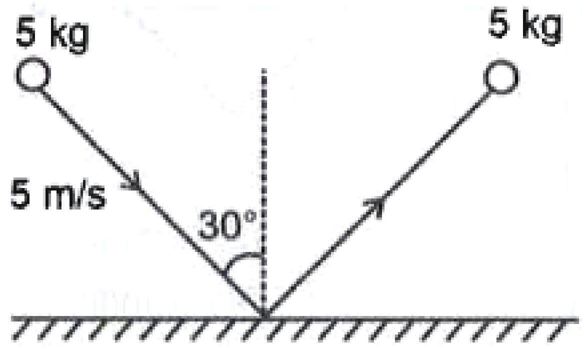
Space for Rough Work

42. When a horse starts suddenly, the rider falls backwards due to

(1) Inertia of rest (2) Inertia of motion

(3) Inertia of direction (4) None of these

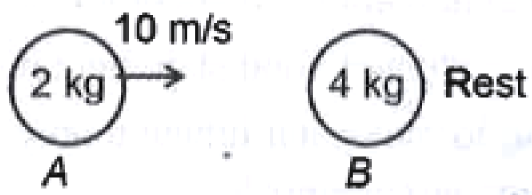
43. A ball of mass 5 kg is projected at angle 30° with the vertical as shown in figure. The angle at which ball will go with respect to vertical after collision will be $\left(e=\frac{1}{2}\right)$



(1) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$ (2) $\tan^{-1}(2)$

(3) $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (4) $\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$

44. Two given masses 2 kg and 4 kg collide elastically (as shown in figure). The loss in kinetic energy during this collision is



(1) 64% (2) 36%

(3) 12% (4) Zero

45. **Assertion (A) :** The scalar product of two vectors can be interpreted as the product of magnitude of the one vector and component of the other vector along the first vector.

Reason (R): Scalar product obeys the commutative and distributive laws.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true and R is NOT the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

46. Consider the following statements.

Statement A : Power of machine gun is determined by both, the number of bullet fired per second and kinetic energy of bullets.

Statement B : Power of any machine is defined as work done by it per unit time.

Based upon above information pick the correct option.

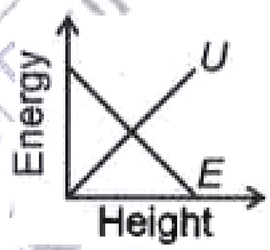
(1) Statement (A) is true but (B) is false

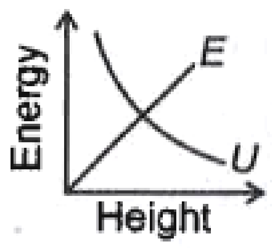
(2) Statement (A) is false but (B) is true

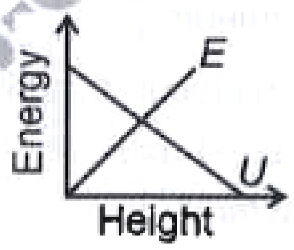
(3) Both statements are true

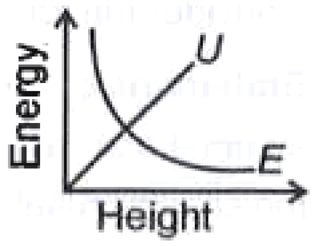
(4) Both statements are false

47. A particle is dropped from the top of a tower of certain height. If E denotes the kinetic energy, U denotes the potential energy, then the curve which shows most appropriate variation of U and E with height will be

(1) 

(2) 

(3) 

(4) 

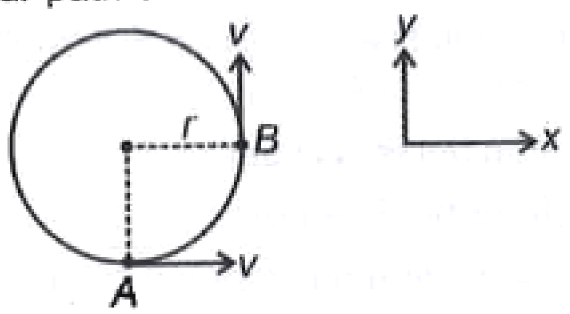
48. The coefficient of restitution e for a perfectly inelastic collision is

(1) 1 (2) 0

(3) 0.5 (4) –1

Space for Rough Work

49. Particle of mass m moves with constant speed v on a circular path of radius r as shown.



The magnitude of change in linear momentum of the particle during its motion from A to B is

(1) $\sqrt{2}mv$

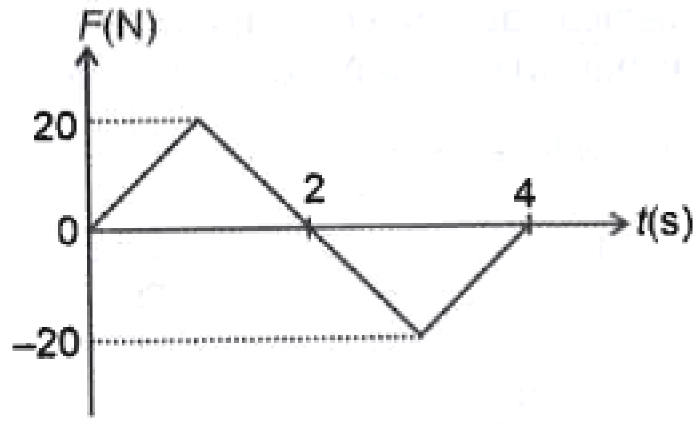
(2) Zero

(3) $\frac{mv}{\sqrt{2}}$

(4) $\sqrt{3}mv$

50. A force (F) acting on a body varies with time (t) as shown in the graph.

If the body is initially at rest, then the velocity acquired by the body at the end of four seconds will be



(1) 5 m/s
(2) 6 m/s
(3) Zero
(4) 2 m/s

[CHEMISTRY]

SECTION-A

51. During change of N_2 to N_2^- the electron is added to which of the following orbitals?

(1) σ^*

(2) σ

(3) π

(4) π^*

52. Consider the following statements.

Statement-I: Sigma (σ) molecular orbitals are symmetrical around the bond-axis while pi (π) molecular orbitals are not symmetrical.

Statement-II: $\sigma 1s$ orbital and $\sigma^* 1s$ orbital do not contain any node.

In light of above statements, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

53. According to molecular orbital theory, both σ-bond and π-bond are present in

(1) B_2
(2) H_2
(3) N_2
(4) C_2

54. Correct order of bond angles in the following molecules is

(1) $BCl_3 > CO_2 > H_2O > NH_3$

(2) $CO_2 > BCl_3 > NH_3 > H_2O$

(3) $CO_2 > NH_3 > BCl_3 > H_2O$

(4) $BCl_3 > NH_3 > H_2O > CO_2$

Space for Rough Work

55. Match list I with list II

	List-I		List-II
a.	NO	(i)	Acidic
b.	Al_2O_3	(ii)	Basic
c.	Na_2O	(iii)	Neutral
d.	SO_2	(iv)	Amphoteric

The correct match is

(1) a(iv), b(iii), c(ii), d(i) (2) a(ii), b(iii), c(i), d(iv)

(3) a(i), b(iv), c(ii), d(iii) (4) a(iii), b(iv), c(ii), d(i)

56. Correct order of ionic radii of the given ions is

(1) $Na^+ > Li^+ > Mg^{2+} > Be^{2+}$

(2) $Li^+ > Na^+ > Mg^{2+} > Be^{2+}$

(3) $Na^+ > Mg^{2+} > Be^{2+} > Li^+$

(4) $Mg^{2+} > Na^+ > Li^+ > Be^{2+}$

57. Correct bond order of the given species is

(1) $O_2 > O_2^- > O_2^+$ (2) $O_2^+ > O_2^- > O_2$

(3) $O_2^+ > O_2 > O_2^-$ (4) $O_2 > O_2^+ > O_2^-$

58. Consider the following statements about PCl_5 molecule

(a) The central atom is sp^3d hybridised.

(b) Axial bonds are stronger than equatorial bonds.

(c) Net dipole moment is zero.

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a), (b) and (c) (4) (a) and (c) only

59. Eka-silicon and Eka-aluminium respectively are

(1) As and Ge

(2) Ca and Ge

(3) Ge and Ga

(4) Ga and Ge

60. Consider the following set of elements

(a) Li, Na, K

(b) Ca, Sr, Ba

(c) N, P, As

Which sets represent Dobereiner's triad?

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

61. Consider the following statements

Statement I : The elements after uranium are called Transuranium elements.

Statement II: Samarium and neptunium are Transuranium elements.

In light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

62. The correct order of dipole moments of the molecules H_2O, NH_3, H_2S and BF_3 is

(1) $H_2O > NH_3 > BF_3 > H_2S$

(2) $H_2O > NH_3 > H_2S > BF_3$

(3) $NH_3 > H_2O > H_2S > BF_3$

(4) $NH_3 > H_2O > BF_3 > H_2S$

63. In which of the following pairs, the two species are iso-structural?

(1) SO_4^{2-} and PO_4^{3-} (2) XeF_4 and SF_4

(3) NO_3^- and SO_3^{2-} (4) BF_3 and NF_3

64. Correct order of ionisation enthalpy of the given alkali metals is

(1) Cs > Rb > K > Na (2) Na > K > Rb > Cs

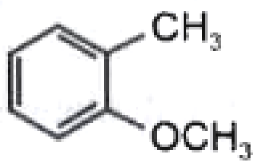
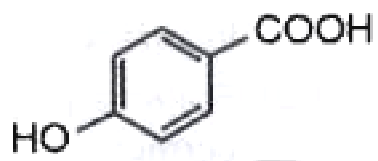
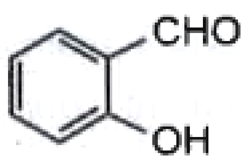
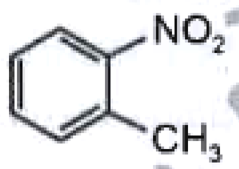
(3) Na > K > Cs > Rb (4) Rb > Cs > Na > K

Space for Rough Work

65. Number of bonding and antibonding electrons present in O_2^- respectively are
(1) 9 and 8 (2) 10 and 7
(3) 12 and 5 (4) 11 and 6

66. Ethyne consists of
(1) 2 sigma and 3 pi bonds
(2) 3 sigma and 1 pi bond
(3) 3 sigma and 2 pi bonds
(4) 1 sigma and 2 pi bonds

67. The pair which contains representative elements is
(1) Zn and Cd (2) Bi and Nd
(3) Ba and Pb (4) Ti and Cu

68. Intramolecular H-bonding is observed in
(1) 
(2) 
(3) 
(4) 

69. Which among the following is a paramagnetic species?
(1) O_2 (2) N_2
(3) C_2 (4) H_2

70. The correct order of decreasing bond length of C–C, C–H, C–O and O–H is
(1) C–C > C–O > C–H > O–H
(2) C–O > C–C > C–H > O–H
(3) C–O > C–C > O–H > C–H
(4) C–C > C–O > O–H > C–H

71. 180° bond angle is absent in
(1) I_3^- (2) CO_2
(3) SF_6 (4) SO_2

72. Total number of lone pairs in XeF_2 is
(1) 6 (2) 4
(3) 10 (4) 9

73. The pair of species which does not exist is
(1) O_2^+ and O_2 (2) He_2^+ and H_2^+
(3) He_2 and Be_2 (4) C_2^+ and B_2

74. The compound which contains both ionic and covalent bonds is
(1) NaBr (2) Na_2SO_4
(3) CCl_4 (4) SO_3

75. Given below are the two statements
Statement I: Bond length is defined as the equilibrium distance between the nuclei of two bonded atoms in a molecule.
Statement II: The covalent radius represents the overall size of the atom which includes its valence shell in a nonbonded situation.
In light of above statements, choose the correct answer.
(1) Statement I is correct but statement II is incorrect
(2) Statement I is incorrect but statement II is correct
(3) Both statement I and statement II are correct
(4) Both statement I and statement II are incorrect

76. Correct order of atomic radii of the given elements is
(1) Al > Si > S > P (2) Al > Si > P > S
(3) Si > Al > P > S (4) S > P > Si > Al

77. Consider the following statements
a. Selenium and sulphur belong to chalcogen group.
b. Antimony and tellurium are metalloids.
c. Rubidium and indium are *s*-block elements.
The correct statements are
(1) a and b only (2) b and c only
(3) a and c only (4) a, b and c

Space for Rough Work

78. Given below are the two statements

Statement I: Ethyl alcohol dissolves in water.

Statement II: Ethyl alcohol forms intermolecular hydrogen bond with water.

In light of above statement, choose the correct option.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

79. C–O bond order in CO_3^{2-} ion is

(1) 1.66 (2) 1.50

(3) 2 (4) 1.33

80. Match list-I with list-II

	List-I Atomic number		List-II IUPAC official name
a.	107	(i)	Rutherfordium
b.	103	(ii)	Bohrium
c.	105	(iii)	Lawrencium
d.	104	(iv)	Dubnium

The correct match is

(1) a(ii), b(iii), c(iv), d(i) (2) a(iii), b(ii), c(iv), d(i)

(3) a(ii), b(iii), c(i), d(iv) (4) a(iv), b(iii), c(ii), d(i)

81. Octet rule is applicable to which pair of compounds?

(1) NO_2 and CH_4 (2) N_2 and CO_2

(3) PF_5 and SF_6 (4) NO and H_2SO_4

82. Given below are the two statements

Statement I: Cl_2O_7 is an acidic oxide.

Statement II: Cl_2O_7 on reaction with water forms $HClO_3$.

In light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

83. Correct decreasing order of negative electron gain enthalpy of the given elements is

(1) S > O > Na > Li (2) O > S > Li > Na

(3) S > O > Li > Na (4) S > Li > O > Na

84. Correct order of electronegativity of the given elements is

(1) C > B > Si > Al (2) B > C > Si > Al

(3) C > Si > B > Al (4) B > Si > C > Al

85. The correct shape and hybridisation for IF_7 are

(1) Trigonal bipyramidal and sp^3d^3

(2) Pentagonal bipyramidal and sp^3d^2

(3) Distorted octahedral and sp^3d^2

(4) Pentagonal bipyramidal and sp^3d^3

SECTION-B

86. Consider the following statements about Fajans rule

(a) The smaller is the size of cation and the larger the size of the anion, the greater the covalent character of an ionic bond.

(b) The greater the charge on the cation, the greater the covalent character of the ionic bond.

(c) For the cations of the same size and charge, the one, with electronic configuration $(n-1)d^{10}ns^0$ is less polarising than the one with noble gas configuration, ns^2np^6

The correct statements are

(1) (a) and (c) only (2) (b) and (c) only

(3) (a), (b) and (c) (4) (a) and (b) only

Space for Rough Work

87. Given below are the two statements

Statement I : O_3 molecule is a resonance hybrid of three canonical structures.

Statement II : Resonance stabilizes the molecule as the energy of the resonance hybrid is less than the energy of any single canonical structure.

In light of above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

88. Match List I with List II

	List I (Species)		List II (Lone pair(s) on central atom)
a.	XeF_6	(i)	2
b.	$[ICl_4]^-$	(ii)	1
c.	CCl_4	(iii)	3
d.	$[I_3]^-$	(iv)	0

The correct match is

(1) a(iv), b(i), c(ii), d(iii) (2) a(ii), b(i), c(iii), d(iv)

(3) a(i), b(ii), c(iv), d(iii) (4) a(ii), b(i), c(iv), d(iii)

89. Which among the following is an endothermic process?

(1) $Ne \longrightarrow Ne^-$ (2) $Br \longrightarrow Br^-$

(3) $Rb \longrightarrow Rb^-$ (4) $Te \longrightarrow Te^-$

90. Highest occupied molecular orbital of O_2^- ion is

(1) π^* (2) σ^*

(3) σ (4) π

91. If the observed dipole moment of A–B molecule is 0.8 D and A–B bond length is 100 pm then percentage ionic character in A–B molecule will be

(1) 25.75% (2) 16.65%

(3) 42.25% (4) 31.85%

92. Which among the following elements is expected to form largest ion to achieve the nearest noble gas configuration?

(1) Mg (2) O

(3) Na (4) F

93. Consider the following statements about SF_6 molecule

(a) Maximum number of atoms lie in a plane are 4

(b) It has regular octahedral geometry

(c) 90° and 180° bond angles are present in the molecule

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a), (b) and (c) (4) (a) and (c) only

94. Given below are the two statements

Statement I: Electronegativity is a qualitative measure of the ability of an atom in a chemical compound to attract shared electrons to itself.

Statement II: Sulphur is the most electronegative element of the third period in periodic table.

In light of above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

Space for Rough Work

95. Match List-I with List-II

	List I		List II
a.	Vanadium	(i)	Halogen
b.	Iodine	(ii)	Lanthanoid
c.	Cerium	(iii)	Actinoid
d.	Thorium	(iv)	Transition metal

The correct match is

(1) a(iii), b(i), c(iv), d(ii)

(2) a(iv), b(i), c(ii), d(iii)

(3) a(ii), b(i), c(iv), d(iii)

(4) a(iii), b(ii), c(i), d(iv)

96. $p\pi$-$d\pi$ bond is absent in

(1) ClO_4^- (2) XeO_3

(3) CO_3^{2-} (4) PO_4^{3-}

97. The pair of elements which show diagonal relationship with each other are

a. Li and Mg b. Be and Al

c. B and P

The correct option is

(1) a and b only (2) b and c only

(3) a and c only (4) a, b and c

98. Intermolecular hydrogen bond is absent in

(1) H_2O (2) NH_3

(3) H_2S (4) HF

99. Consider the following statements.

a. In OF_2 the oxidation state of oxygen is +1

b. Maximum covalency of boron in its compounds is 4

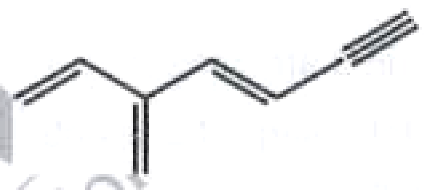
c. $p_\pi - p_\pi$ bond is present in N_2 molecule

The correct statements are

(1) a and b only (2) b and c only

(3) a and c only (4) a, b and c

100. Total number of π-bond electrons in the following molecule is



(1) 5

(2) 10

(3) 8

(4) 4

[BOTANY]

SECTION-A

101. In the given plants, how many of them are belonging to a family having the floral formula $\oplus ⚥ K_{(5)} \overset{\frown}{C_{(5)} A_5} \underline{G}_{(2)}$

Potato, *Petunia*, Belladonna, Mustard, Sugarcane, Wheat, Maize

(1) 4 (2) 5

(3) 3 (4) 2

102. Select the **incorrectly** matched pair w.r.t. stamens.

(1) Epipetalous – Brinjal

(2) Epiphyllous – Lily

(3) Polyadelphous – *Citrus*

(4) Monoadelphous – Pea

Space for Rough Work

103. Read the given statements (a-d)

a. Ovules develop into seeds after fertilization.

b. The hilum is a scar on the seed coat through which the developing seeds were attached to the fruit.

c. Nucellus remains persistent in some seeds and is called pericarp.

d. The plumule and radicle are enclosed in sheaths which are called as coleoptile and coleorhiza respectively.

In the light of above statements, select the **correct** option.

(1) All are correct except d

(2) Only b and c are correct

(3) Only a, b and d are correct

(4) Only c and d are correct

104. Read the following statements and select the **correct** option.

Statement-A: A sterile stamen is called staminode.

Statement-B: When stamen is attached to the petal, it is called epiphyllous.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

105. Root hairs develop from the

(1) Root cap

(2) Elongation zone

(3) Maturation zone

(4) Region of meristematic activity

106. Select the **incorrectly** matched pair.

(1) Marginal placentation – Pea

(2) Axile placentation – China rose

(3) Basal placentation – *Argemone*

(4) Parietal placentation – Mustard

107. Apocarpous condition is seen in

(1) Lotus and tomato (2) Lotus and mustard

(3) Mustard and tomato (4) Lotus and rose

108. The arrangement of ovules on placenta within the ovary is known as

(1) Placentation (2) Aestivation

(3) Inflorescence (4) Venation

109. Flowers are borne in an acropetal succession in all of the following plants, **except**

(1) Radish (2) Mustard

(3) Lupin (4) *Dianthus*

110. Which of the following is usually **not** the function of root?

(1) Synthesis of plant growth regulators

(2) Absorption of water and minerals from soil

(3) Storage of reserve food materials

(4) Manufacturing of food and its transportation to aerial parts of plant

111. Read the following Assertion (A) and Reason (R) and select the **correct** option.

Assertion (A): In cymose inflorescence, the growth of main axis is limited.

Reason (R): In cymose inflorescence, the main axis terminates into a flower.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

Space for Rough Work

112. Match column-I with column-II and select the **correct** option w.r.t. aestivation in corolla.

	Column-I		Column-II
a.	Valvate	(i)	Pea
b.	Twisted	(ii)	*Cassia*
c.	Imbricate	(iii)	*Calotropis*
d.	Vexillary	(iv)	Cotton

(1) a(iii), b(iv), c(ii), d(i) (2) a(iv), b(iii), c(i), d(ii)
(3) a(iv), b(iii), c(ii), d(i) (4) a(iii), b(iv), c(i), d(ii)

113. Select the **odd** one out w.r.t. radial symmetry.
(1) Mustard (2) *Datura*
(3) Chilli (4) Bean

114. Read the given statements from (a-d)
a. Perigynous flower has half inferior ovary.
b. Epigynous flower has inferior ovary.
c. Hypogynous flower has superior ovary.
d. In mustard and China rose, gynoecium is situated in the centre and other parts of the flower are located on the rim or periphery of the thalamus.

In the light of above statements select the **correct** option.
(1) a, b and d are correct
(2) Only a and b are correct
(3) All are correct except d
(4) Only c and d are correct

115. Read the following statements and state them as **true(T)** or **false(F)** and select the **correct** option.
a. Anthers are usually bilobed and each lobe contains two pollen sacs.
b. Corolla is the second whorl of the flower which is generally brightly coloured and have fragrance.
c. Sepals are generally green leaf-like structure that protect the flower in bud stage.

	a	b	c
(1)	T	F	F
(2)	F	T	T
(3)	T	T	T
(4)	F	T	F

116. Select the **mismatched** pair.
(1) Viroids – Potato spindle tuber disease
(2) Prions – Mad cow disease in cattle
(3) *Alternaria* – Late blight of potato
(4) *Claviceps* – Ergot disease

117. Read the following statements and select the **correct** option.

Statement-A: In a symbiotic association between algae and fungi, algae prepare food for fungi and fungi provide shelter and absorb mineral nutrients and water for its partner.

Statement-B: Lichens are very good pollution indicators.

(1) Only statement A is correct
(2) Only statement B is correct
(3) Both the statements A and B are correct
(4) Both the statements A and B are incorrect

118. The causative agents of Cr-Jacob disease in humans are
(1) Viruses (2) Viroids
(3) Prions (4) Bacteria

119. Which of the following characteristics of euglenoids is similar to plants?
(1) Presence of protein rich layer called pellicle
(2) Presence of photosynthetic pigments
(3) Presence of contractile vacuole
(4) Formation of non-motile gametes during sexual reproduction

Space for Rough Work

120. Mycorrhiza is **correctly** described as
(1) Parasitic association between root and fungi
(2) Symbiotic association between algae and fungi
(3) Symbiotic association between roots of higher plants and fungi
(4) Parasitic association between algae and fungi

121. Specialised cells called heterocysts are present in
(1) Slime moulds (2) Blue green algae
(3) Protists (4) Golden algae

122. Select the **incorrectly** matched pair.
(1) Amoeboid protozoan – *Amoeba*
(2) Flagellated protozoan – *Trypanosoma*
(3) Ciliated protozoan – *Entamoeba*
(4) Sporozoan – *Plasmodium*

123. Cell wall of fungi is composed of
(1) Chitin and polysaccharides
(2) Peptidoglycan
(3) Cellulose only
(4) Murein

124. Read the following statements and state them as true (**T**) or false (**F**) and select the **correct** option.
a. Diatoms float passively in water currents.
b. Reserve food material is stored in the form of oil and glycogen in fungi.
c. Cyanobacteria are characterised by the presence of flagellum throughout the life cycle.
d. Kingdom fungi is classified only on the basis of fruiting bodies into various classes.
e. Deuteromycetes are commonly known as imperfect fungi.

	a	b	c	d	e
(1)	T	T	F	F	T
(2)	F	F	T	T	F
(3)	T	F	T	F	F
(4)	F	T	T	F	T

125. Which of the following statements is **not** true w.r.t. slime moulds?
(1) During unfavourable conditions, the plasmodium differentiates and forms fruiting bodies
(2) The spores possess true walls and are extremely resistant
(3) The spores are dispersed by air currents
(4) Slime moulds are the only parasitic protists

126. Bacterial viruses
a. Infect the sole members of kingdom Monera
b. Usually have double stranded DNA
c. Are also called bacteriophage
d. Generally lack protein coat

Choose the **correct** option.
(1) Only a is correct
(2) Only a and c are correct
(3) Only b is correct
(4) Only d is incorrect

127. Read the given statements (a-d).
a. Methanogens are present in the gut of several ruminant animals.
b. Chemosynthetic autotrophic bacteria oxidise nitrates, nitrites and ammonia and use the released energy for their ATP production.
c. Cyanobacteria often form blooms in polluted water bodies.
d. Thermoacidophiles are found in extreme salty areas and cannot survive in water having low pH.

In the light of above statements, select the **correct** option.
(1) All are correct except d
(2) Only a and b are correct
(3) Only a and d are correct
(4) Only b and c are correct

Space for Rough Work

128. Match column-I with column-II and select the **correct** option.

	Column-I		Column-II
a.	*Rhizopus*	(i)	Mushroom
b.	*Ustilago*	(ii)	Rust fungus
c.	*Puccinia*	(iii)	Smut fungus
d.	*Agaricus*	(iv)	Bread mould

(1) a(iv), b(iii), c(ii), d(i) (2) a(iv), b(iii), c(i), d(ii)

(3) a(iii), b(iv), c(ii), d(i) (4) a(iii), b(iv), c(i), d(ii)

129. Identify the **incorrect** statement w.r.t. Archaebacteria.

(1) They differ from other bacteria in having a different cell wall structure

(2) Cell wall contains branched chain lipids

(3) They have introns in their genetic sequence

(4) They can also survive in most harsh habitats

130. Select the **odd** one out w.r.t. bacterial diseases.

(1) Cholera (2) Typhoid

(3) Tetanus (4) Sleeping sickness

131. Select the **odd** one out for mango fruit.

(1) One seeded

(2) Develops from monocarpellary ovary

(3) Has fleshy, juicy mesocarp

(4) Epicarp is fibrous edible part

132. Which of the following statements is **incorrect** w.r.t. Mycoplasma?

(1) They completely lack cell wall

(2) They are pathogenic in animals and plants

(3) They cannot survive without oxygen

(4) They are insensitive to penicillin

133. Select the **mismatched** pair.

(1) Coccus – Spherical

(2) Bacillus – Rod shaped

(3) Vibrium – Irregular shape

(4) Spirillum – Spiral

134. Identify the **incorrect** statement w.r.t. characteristics of the members of Monera.

(1) They do not have nuclear membrane, nucleolus and chromatin associated with histone proteins

(2) Ribosomes are made up of two subunits 50S and 30S

(3) Some of the bacteria are heterotrophic and vast majority are autotrophic

(4) Respiratory enzymes are found associated with plasma membrane

135. In a leguminous plant, swollen leaf base is called

(1) Stipule (2) Pulvinus

(3) Midrib (4) Lamina

SECTION-B

136. Select the **incorrectly** matched pair.

(1) Endospermic seed – Castor

(2) Non-endospermic seed – Gram

(3) Mesocarp – Fleshy and edible in coconut

(4) Endocarp – Hard and stony in mango

137. Which of the given regions/structures of root tip protects the tender apex of the root as it makes its way through the soil?

(1) Root cap

(2) Region of elongation

(3) Region of maturation

(4) Region of meristematic activity

Space for Rough Work

138. Read the following characteristics (a-d):
 a. Racemose type of inflorescence is present
 b. Flowers are bisexual, usually actinomorphic but sometimes zygomorphic
 c. Siliqua or silicula type of fruit are seen
 d. Sepals and petals are four in number with imbricate and valvate aestivations, respectively

 These above characteristics are **true** for members of

 (1) Brassicaceae (2) Compositae
 (3) Poaceae (4) Malvaceae

139. Identify the **incorrect** statement.
 (1) Leaflets are attached at a common point, *i.e.*, at the tip of petiole in silk cotton
 (2) In *Salvia*, there may be variation in length of filaments within a flower.
 (3) Rachis represents the midrib of the leaf in palmately compound leaf
 (4) In some plants like *Monstera* and banyan tree, roots arise from parts of plant other than radicle

140. Which of the following pairs of fungi are edible and considered delicacies?
 (1) Morels and *Alternaria* (2) *Albugo* and *Claviceps*
 (3) Morels and Truffles (4) *Yeast* and *Agaricus*

141. Leaves of all the following plants show reticulate venation, **except**
 (1) Peepal (2) *Hibiscus*
 (3) *Luffa* (4) Banana

142. Read the following Assertion (A) and Reason (R) statements and select the **correct** option.

 Assertion (A): Viruses can pass through bacteria proof filters.

 Reason (R): Viruses are exception to the cell theory and are obligatory intracellular parasites.

 (1) (A) is true but (R) is false
 (2) Both (A) and (R) are false
 (3) Both (A) and (R) are true and (R) is the correct explanation of (A)
 (4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

143. Identify the **incorrect** statement w.r.t. dinoflagellates.
 (1) These organisms are mostly marine and heterotrophic
 (2) The cell wall has stiff cellulosic plates on the outer surface
 (3) Most of them have two flagella
 (4) *Gonyaulax* is responsible for red tides

144. Which of the following fungi is used extensively in biochemical and genetic work?
 (1) *Aspergillus*
 (2) *Neurospora*
 (3) *Trichoderma*
 (4) Yeast

145. Viruses could be crystallised, was shown by
 (1) T.O. Diener (2) M.W. Beijerinck
 (3) W.M. Stanley (4) D.J. Ivanowsky

146. Three-domain system of classification was proposed by __A__, that divides the kingdom Monera into __B__ domains.

 Choose the option that **correctly** fills the blanks A and B.

	A	B
(1)	Carl Woese	Two
(2)	R.H. Whittaker	One
(3)	Aristotle	Two
(4)	Linnaeus	Three

Space for Rough Work

147. Which one of the following statements is **incorrect** w.r.t. basidiomycetes?
(1) Karyogamy and meiosis occurs in basidium
(2) Sexual reproduction does not involve sex-organs
(3) Mycelium is branched and septate
(4) Secondary mycelium is short lived but dominant phase of life cycle

148. Symbols used in floral formula for polysepalous and gamosepalous condition respectively are
(1) K_n and $K_{(n)}$
(2) $K_{(n)}$ and K_n
(3) C_n and $C_{(n)}$
(4) $C_{(n)}$ and C_n

149. Match column-I with column-II and select the **correct** option.

	Column-I		Column-II
a.	Malvaceae	(i)	Br%♀$K_{\text{pappus or }0}C_{(5)}A_0\bar{G}_{(2)}$
b.	Gramineae	(ii)	Ebr⊕or%⚥$K_{2+2}C_{\times 4}A_{2+4}\underline{G}_{(2)}$
c.	Cruciferae	(iii)	%⚥$P_{\text{0 or 2 or 3(Lodicules)}}A_{\text{3 or 6}}\underline{G}_1$
d.	Compositae	(iv)	Epi_{5-7}⊕⚥$K_{(5)}\widehat{C_5 A}_{(\infty)}\underline{G}_{(5)}$

(1) a(iii), b(iv), c(i), d(ii) (2) a(iii), b(iv), c(ii), d(i)
(3) a(ii), b(iv), c(iii), d(i) (4) a(iv), b(iii), c(ii), d(i)

150. Select the **odd** one out w.r.t. alternate phyllotaxy.
(1) Sunflower (2) China rose
(3) Guava (4) Mustard

[ZOOLOGY]

SECTION-A

151. In the human respiratory tract, the primary site for exchange of gases is lined by
(1) Simple cuboidal epithelium
(2) Simple columnar epithelium
(3) Simple squamous epithelium
(4) Compound epithelium

152. In a standard ECG, the P-wave represents the
(1) Electrical excitation of the atria
(2) End of ventricular systole
(3) Repolarisation of the lower chambers of the heart
(4) Initiation of depolarisation of the ventricles

153. The hepatic portal vein carries blood from
(1) Liver to the kidney (2) Kidney to the liver
(3) Liver to the intestine (4) Intestine to the liver

154. Partial pressure of a gas
(1) Is the pressure contributed by an individual gas in a mixture of gases
(2) Is the pressure contributed by all gases in a mixture of gases
(3) Cannot affect the rate of diffusion of gases
(4) Present in atmosphere is equal to the atmospheric pressure

155. Select the **incorrect** match.
(1) Fibrinogens – Help in coagulation of blood
(2) Globulins – Facilitate defense mechanism
(3) Albumins – Maintain osmotic balance
(4) Serum – Plasma with clotting factors

156. All of the following are correct w.r.t. human RBCs, **except**
(1) Formed in the red bone marrow in the adults
(2) Devoid of nucleus
(3) Contain red-coloured, iron containing complex protein
(4) Biconvex in shape

Space for Rough Work

157. Read the following statements.

Statement A: Spleen is called the graveyard of RBCs.

Statement B: In a standard ECG, the number of QRS complexes that occur in a given time period can be counted to determine the stroke volume directly.

Choose the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

158. Purkinje fibres present in the human heart are the modification of

(1) Nervous tissue

(2) Muscle tissue

(3) Dense connective tissue

(4) Specialised connective tissue

159. Select the **incorrect** statement w.r.t. the human heart.

(1) It has the size of a clenched fist.

(2) It is a mesodermally derived structure.

(3) Its right ventricle has the thickest wall.

(4) It is covered by a double layered pericardium.

160. If repeated checks of blood pressure of an individual shows systolic and diastolic blood pressure equal to or higher than ______ and ______ respectively, it indicates hypertension.

Select the **correct** option to fill in the blanks.

(1) 110; 80 mm Hg

(2) 100; 80 mm Hg

(3) 120; 70 mm Hg

(4) 140; 90 mm Hg

161. Match column I with column II.

	Column I		Column II
a.	Heart attack	(i)	Heart stops beating
b.	Cardiac arrest	(ii)	Not enough O_2 is reaching heart muscles that leads to chest pain
c.	Heart failure	(iii)	Heart is not pumping blood effectively enough to meet the body's demand
		(iv)	Heart muscles are suddenly damaged by an inadequate blood supply

Select the **correct** option.

(1) a(iv), b(i), c(iii)

(2) a(i), b(ii), c(iii)

(3) a(iii), b(ii), c(i)

(4) a(i), b(iii), c(ii)

162. The opening of the right and left ventricle into pulmonary artery and the aorta respectively are guarded by the

(1) Mitral valve

(2) Semilunar valve

(3) Tricuspid valve

(4) Bicuspid valve

163. In a standard ECG, return of ventricles to the normal state from excited state is represented by

(1) T-wave

(2) QRS complex

(3) P-wave

(4) Q-wave

164. In a healthy adult human, an average life span of the most abundant formed element of blood is

(1) 100 days

(2) 120 days

(3) 36 days

(4) 48 days

Space for Rough Work

165. Read the following statements w.r.t. humans.

Statement A: Upon release of adrenal medullary hormones, cardiac output increases.

Statement B: If the parasympathetic influence on heart is removed, heart will beat slowly.

Select the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

166. Exchange of a gas at the diffusion membrane in human respiratory system is mainly based on all, **except**

(1) Concentration gradient of the gas

(2) Thickness of the membrane

(3) Solubility of the gas

(4) Total pressure of all the gases present in alveoli

167. In a healthy human, normal pumping pressure and resting pressure is respectively

(1) 80 mm Hg; 120 mm Hg

(2) 120 mm Hg; 80 mm Hg

(3) 100 mm Hg; 90 mm Hg

(4) 80 mm Hg; 100 mm Hg

168. Select the **incorrect** match w.r.t. blood vessels.

(1)	Tunica intima	–	An inner lining of simple squamous epithelium
(2)	Tunica media	–	A middle layer of skeletal muscle and elastic fibres
(3)	Tunica externa	–	An external layer of fibrous connective tissue with collagen fibres
(4)	Tunica media	–	Thinner in veins as compared to arteries

169. **Assertion (A):** Neural signals from pneumotaxic centre can reduce the duration of inspiration.

Reason (R): Pneumotaxic centre present in the medulla region of brain can moderate the functions of the respiratory rhythm centre.

In the light of above statements, select the **most appropriate** option.

(1) Both (A) and (R) are true; (R) is the correct explanation of (A)

(2) Both (A) and (R) are true; (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

170. Pulmonary ventilation is the process in which

(1) Gases diffuse across alveolar membrane.

(2) Atmospheric air is drawn in and CO_2 rich alveolar air is released out.

(3) O_2 is utilised by the cells for catabolic reactions and resultant release of CO_2.

(4) Gases diffuse between blood and tissues.

171. A healthy adult human breathes 'X' times/minute under normal condition. 'X' is numerically equal to

(1) % of protein present in blood plasma

(2) Amount of haemoglobin present in every 100 mL of blood of a healthy adult human

(3) % of monocytes in total leucocyte count

(4) Number of heart beats per minute of a healthy man

172. Read the following statements.

a. In reptiles, the left atrium receives deoxygenated blood from lungs.

b. The cardiac output of an adult healthy man is 5 L

c. Sympathetic nerve fibres can increase the strength of ventricular contraction in humans.

Select the **correct** option.

(1) a, b and c are correct

(2) a, b and c are incorrect

(3) b and c are correct

(4) Only b is correct

Space for Rough Work

173. A person met with an accident and needed immediate blood transfusion but his blood group was unknown to his family members. In this case, which of the following blood groups can be used, so that no mismatching occurs?
(1) AB –ve (2) O +ve
(3) O –ve (4) AB +ve

174. In humans, the first heart sound is due to
(1) Closure of valves which prevent back flow of blood into atria from ventricles
(2) Opening of valves that is formed of three muscular flaps
(3) Closure of valves which prevent back flow of blood into ventricles from aorta
(4) Opening of mitral valve

175. How many of the structures given in the box below help in transporting the atmospheric air to alveoli?

Terminal bronchioles, External nostrils, Trachea, Primary bronchi, Nasal chamber

Select the **correct** option.
(1) Five (2) Three
(3) Four (4) Two

176. Maximum volume of air one can breathe in after a forceful expiration is
(1) Functional residual capacity
(2) Inspiratory reserve volume
(3) Total lung capacity
(4) Vital capacity

177. Rh incompatibility can be observed in between the
(1) Rh +ve pregnant mother with Rh –ve foetus
(2) Rh –ve pregnant mother with Rh +ve foetus
(3) Rh –ve pregnant mother with Rh –ve foetus
(4) Rh +ve pregnant mother with Rh +ve foetus

178. Read the following statements.
Statement A: Aquatic molluscs exhibit pulmonary respiration.
Statement B: Humans have the ability to increase the strength of inspiration and expiration with the help of additional muscles in the abdomen.

Select the **correct** option.
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

179. When percentage saturation of haemoglobin with O_2 on y-axis is plotted against the pO_2 on x-axis in a graph, the curve obtained is
(1) Parabola (2) Sigmoid
(3) Hyperbola (4) Linear

180. A chemosensitive area situated adjacent to the respiratory rhythm centre is highly sensitive to
(1) Rise in H^+ and pO_2 (2) Fall in H^+ and pO_2
(3) Rise in pCO_2 and H^+ (4) Fall in pCO_2 and pO_2

181. What is the volume of air inspired or expired by a healthy adult man, during a normal respiration?
(1) 1000 mL (2) 8000 mL
(3) 500 mL (4) 1500 mL

182. The enzyme 'E' which catalyses the given reaction is present in

$$CO_2 + H_2O \xrightleftharpoons{\text{'E'}} H_2CO_3 \xrightleftharpoons{\text{'E'}} HCO_3^- + H^+$$

(1) Platelets (2) Lymphocytes
(3) Erythrocytes (4) Monocytes

183. Select the **correct** statement w.r.t. humans.
(1) The partial pressure of 'O_2' is 45 mm Hg in blood vessel carrying deoxygenated blood.
(2) The partial pressure of 'O_2' is 40 mm Hg in alveoli
(3) The partial pressure of 'O_2' is more in deoxygenated blood than oxygenated blood
(4) The partial pressure of 'O_2' is 95 mm Hg in systemic arteries that carry oxygenated blood

184. In a healthy man, the volume of air that will remain in the lungs after a normal expiration is
(1) IC + EC – TV (2) TLC – IC
(3) TLC – FRC (4) IC + FRC

Space for Rough Work

185. Which of the following is an **incorrect** statement w.r.t. cardiac cycle in a normal healthy adult human?
(1) During joint diastole, bicuspid valve remains open.
(2) During ventricular diastole, the ventricular pressure rises causing the closure of semilunar valves.
(3) Atrial diastole coincides with the ventricular systole.
(4) It consists of joint diastole, atrial systole and ventricular systole.

SECTION-B

186. Every 500 mL of blood found in the pulmonary vein can deliver around how many mL of oxygen to the tissues under normal physiological conditions?
(1) 5 mL
(2) 10 mL
(3) 20 mL
(4) 25 mL

187. Select the **correct** match w.r.t. respiratory volumes/capacities in an adult human.
(1) IC – 1100-1200 mL
(2) FRC – 5400-5600 mL
(3) RV – 1500-1600 mL
(4) IRV – 2500-3000 mL

188. Select the correct set of factors that favour the formation of oxyhaemoglobin.
(1) Low pO_2, high H^+ concentration, low pCO_2
(2) High pCO_2, high temperature, low pH
(3) High pO_2, low pCO_2, high pH
(4) Low pO_2, high pCO_2, high temperature

189. Which of the following statements is **correct** w.r.t. humans?
(1) A special neural centre in the medulla oblongata can moderate the cardiac functions through somatic nervous system.
(2) A special coronary system of blood vessels is present exclusively for the circulation of blood to and from the cardiac musculature.
(3) The pulmonary circulation provides nutrients, O_2 and other essential substances to all the body tissues and takes CO_2 and other harmful substances away for elimination.
(4) Silicosis is not an occupational respiratory disorder.

190. Select the **incorrect** match w.r.t. respiratory structures of different organisms.
(1) Earthworm – Moist cuticle
(2) Insect – Tracheal tubes
(3) Aquatic arthropod – Skin
(4) Reptile – Lungs

191. Select the **incorrect** statement w.r.t. lymph.
(1) It has the same mineral distribution as that in plasma.
(2) It is a colourless fluid devoid of lymphocytes.
(3) Fats are absorbed through it in the lacteals present in the intestinal villi.
(4) It is an important carrier for nutrients, hormones, *etc.*

192. How many of the following is/are involved in coagulation of blood?

Calcium ions, Prothrombins, Platelet factors, Fibrin

Select the **correct** option.
(1) Four (2) Two
(3) Three (4) One

Space for Rough Work

193. In humans, all of the following constitute the respiratory diffusion membrane, **except**
   (1) Cellular basement membrane
   (2) Endothelium of alveolar capillaries
   (3) Thin squamous epithelium of alveoli
   (4) Acellular basement membrane

194. During normal inspiration
   (1) Contraction of internal inter-costal muscles lifts up the ribs and sternum
   (2) Contraction of diaphragm increases the volume of thoracic chamber in dorso-ventral axis
   (3) Increase in pulmonary volume increases the intra-pulmonary pressure
   (4) Increase in the thoracic volume causes increase in the pulmonary volume

195. Read the following statements.

   Statement A: The anatomical set up of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the pulmonary cavity.

   Statement B: In humans, the outer pleural membrane of lungs is in close contact with the lung surface.

   Select the **correct** option.
   (1) Both statements A and B are correct
   (2) Both statements A and B are incorrect
   (3) Only statement A is correct
   (4) Only statement B is correct

196. Match column I with column II and select the **correct** option w.r.t. formed elements and their percentage (in total WBCs) in a healthy man.

	Column I		Column II
a.	Monocytes	(i)	20-25
b.	Neutrophils	(ii)	6-8
c.	Lymphocytes	(iii)	2-3
		(iv)	60-65

   (1) a(i), b(ii), c(iii)
   (2) a(ii), b(iv), c(i)
   (3) a(iii), b(i), c(iv)
   (4) a(iv), b(i), c(iii)

197. Select the correct statement w.r.t. the pacemaker of the human heart.
   (1) It is present in the right lower corner of the right atrium.
   (2) It can generate action potentials only with the stimulus of nerve impulse.
   (3) It can generate the maximum number of action potentials, *i.e.*, 70-75 min^{-1}.
   (4) It is known as atrio-ventricular node.

198. Select the correct set of respiratory volumes/ capacities which cannot be measured by a simple spirometer.
   (1) TV, ERV
   (2) VC, IRV
   (3) FRC, TLC
   (4) EC, IC

199. Two separate blood circulatory pathways are present in
   (1) Birds, mammals
   (2) Earthworms, flatworms
   (3) Sponges, fishes
   (4) Amphibians, reptiles

200. Complete the analogy by selecting the **correct** option w.r.t. thoracic chamber.

   Ventrally : Sternum :: Laterally : _______
   (1) Ribs
   (2) Thoracic vertebrae
   (3) Diaphragm
   (4) Cervical vertebrae

❑ ❑ ❑

Space for Rough Work

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